



MBFI Assignment

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Introduction

The Global Financial Crisis of 2008 hit almost every country hard. None of them anticipated it. It had an enormous impact on Canada even though the GFC originated in USA. It was caused by a number of factors, including the collapse of the subprime mortgage market, the failure of major financial institutions, the freezing of credit markets, ect.

The Canadian economy contracted by 3.3% in 2009, the worst recession since the Great Depression. Unemployment rose to 8.7%, and housing prices fell by 20%.

In this assignment, I have tried to link the factors like 1-year Tbill, Overnight rate (the key interest rate of Bank of Canada), Inflation Rate, and most importantly, "Credit Default Spread" and analyze the data accordingly.

1-Year-T-Bill, Overnight Rate, Inflation Rate, CDS and CDS Spread

1-Year-T-Bill:

A 1-year Treasury bill (T-bill) is a short-term debt security issued by the government of Canada with a maturity of one year. T-bills are sold at a discount to their face value and pay the face value when they mature. For example, if you buy a \$100 T-bill with a 5% interest rate, you will pay \$95 for it. When the T-bill matures in one year, you will receive \$100. This means that you will earn \$5 in interest over the one-year period.

T-bills are considered to be very safe investments because they are backed by the full faith and credit of the Canadian government. This means that if the government defaults on its debt, you will still be able to get your money back.

T-bills are also very liquid, which means that they can be easily bought and sold. This makes them a good investment for investors who need to raise cash quickly. There is also very low risk associated with a T-Bill. To buy a 1-year T-bill, you can go through a brokerage firm or directly through the Bank of Canada.

However, they give low-yield, there is a risk of inflation (inflation can destroy the investment) and also a risk of interest rates going up. Value of a T-bill is also affected by



the prevailing interest rates in the market. When interest rates rise, the value of existing T-bills with lower interest rates decreases. This is because investors are willing to pay less for a T-bill with a lower interest rate when they can get a higher interest rate on a new T-bill.

Overnight Rate:

Overnight rates are the interest rates that banks charge each other for overnight loans. The Bank of Canada (BoC) sets the target for the overnight rate, and this rate influences other interest rates in the economy, such as mortgage rates and credit card rates.

Overnight rates are set 8 times a year, but for simplicity of calculation and data analysis, I have taken the average of those 8 times.

The BoC uses the target for the overnight rate as its main monetary policy tool. By changing the target for the overnight rate, the BoC can influence the amount of money in circulation in the economy. When the BoC raises the target for the overnight rate, it makes it more expensive for banks to borrow money, which reduces the amount of money in circulation. This can help to slow down economic growth and inflation.

When the BoC lowers the target for the overnight rate, it makes it less expensive for banks to borrow money, which increases the amount of money in circulation. This can help to stimulate economic growth.

Inflation Rate:

Inflation rate is the rate at which the prices of goods and services in an economy are rising. It is typically measured as a percentage change in the Consumer Price Index (CPI) over a period of time.

CDS:

In the movie 'The Big Short', CDS were used to short (bet against) MBS when they saw a bubble and anticipated it. A credit default swap (CDS) is a financial instrument that allows one party to transfer the credit risk of a debt obligation to another party. The party that assumes the credit risk is called the protection buyer, and the party that transfers the credit risk is called the protection seller.



The protection buyer makes periodic payments to the protection seller, and in return, the protection seller agrees to pay the protection buyer if the underlying debt obligation defaults. The underlying debt obligation can be a bond, loan, Mortgage backed securities, or other debt instrument.

CDS are often used by investors to hedge against the risk of default. For example, an investor who owns a bond may buy a CDS to protect themselves against the risk of the bond defaulting to burst.

CDS Spread:

A credit default swap (CDS) spread is the premium that a protection buyer pays to a protection seller in exchange for the seller's promise to pay the buyer if the underlying debt obligation defaults. The CDS spread is expressed as a percentage of the face value of the underlying debt obligation.

For example, if the CDS spread on a bond is 100 basis points, the protection buyer would pay the protection seller 0.10% of the face value of the bond each year. If the bond defaults, the protection seller would pay the protection buyer the full face value of the bond.

The CDS spread is determined by a number of factors, including the creditworthiness of the issuer of the underlying debt obligation, the term of the CDS, and the volatility of the underlying debt obligation. A higher credit risk will lead to a higher CDS spread, and a longer term CDS will also lead to a higher CDS spread. A higher volatility of the underlying debt obligation will also lead to a higher CDS spread.

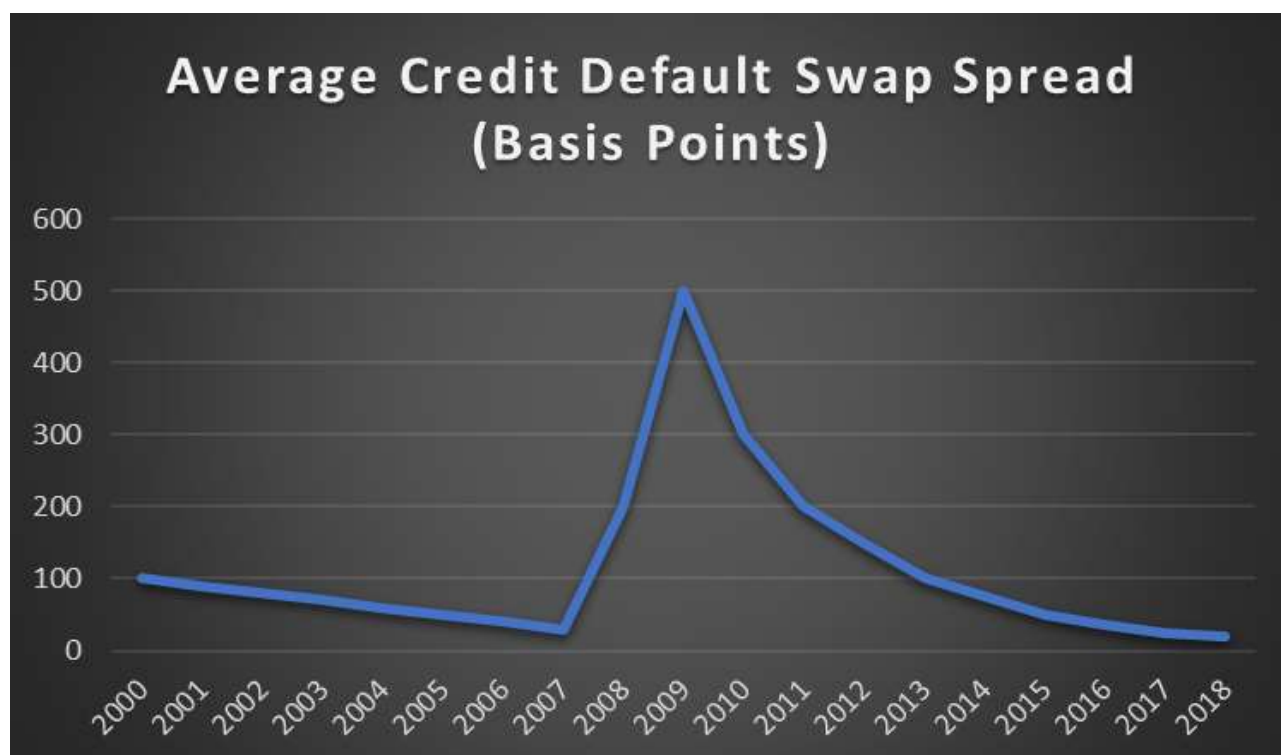
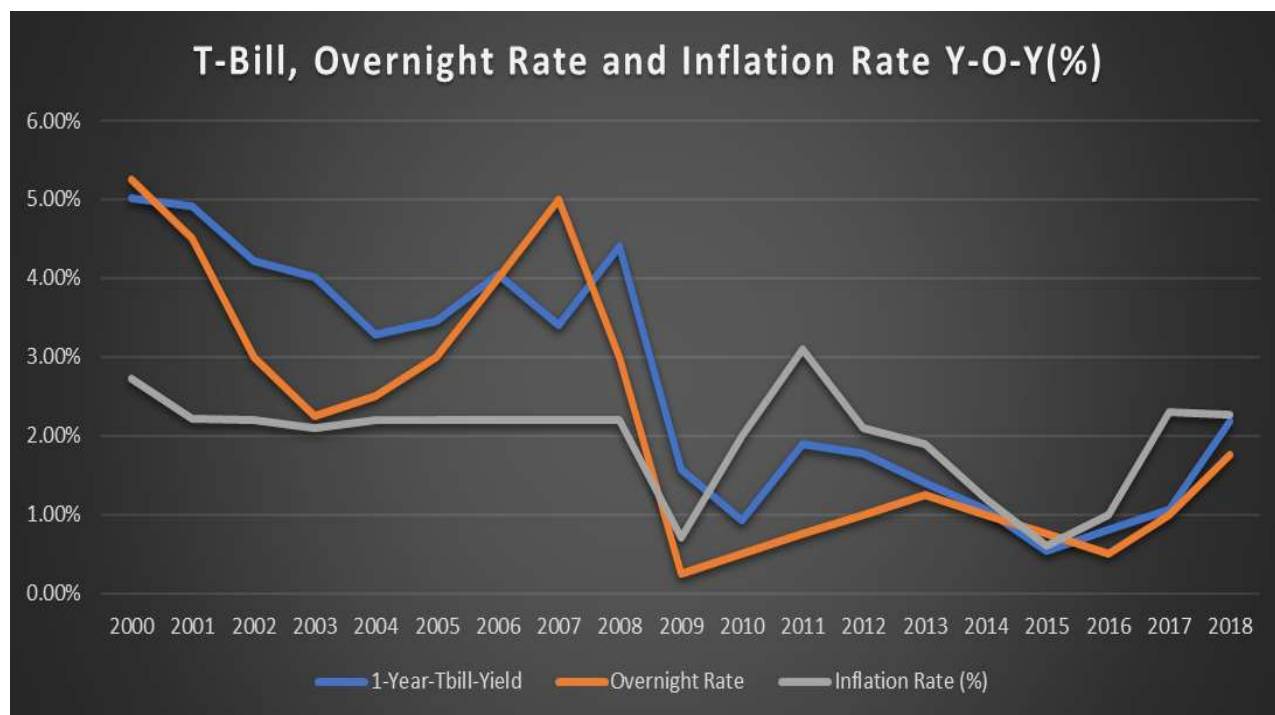
The CDS spread is a useful tool for measuring the credit risk of an underlying debt obligation. A higher CDS spread indicates that the market is more concerned about the risk of default. This can be useful for investors who are trying to assess the risk of their investments.

The CDS spread can also be used to speculate on the creditworthiness of an issuer. If an investor believes that an issuer is likely to default, they may buy a CDS on the issuer's debt. If the issuer does default, the investor will profit from the CDS.

Data

Year	1-Year-Tbill-Yield	Overnight Rate	Inflation Rate (%)	Average Credit Default Swap Spread
2000	5.02%	5.25%	2.72%	100 basis points
2001	4.92%	4.50%	2.22%	90 basis points
2002	4.22%	3.00%	2.20%	80 basis points
2003	4.01%	2.25%	2.10%	70 basis points
2004	3.29%	2.50%	2.20%	60 basis points
2005	3.46%	3.00%	2.20%	50 basis points
2006	4.04%	4.00%	2.20%	40 basis points
2007	3.40%	5.00%	2.20%	30 basis points
2008	4.40%	3.00%	2.20%	200 basis points
2009	1.58%	0.25%	0.70%	500 basis points
2010	0.93%	0.50%	2.00%	300 basis points
2011	1.90%	0.75%	3.10%	200 basis points
2012	1.78%	1.00%	2.10%	150 basis points
2013	1.40%	1.25%	1.90%	100 basis points
2014	1.07%	1.00%	1.20%	75 basis points
2015	0.53%	0.75%	0.60%	50 basis points
2016	0.80%	0.50%	1.00%	35 basis points
2017	1.07%	1.00%	2.30%	25 basis points
2018	2.19%	1.75%	2.27%	20 basis points

Graphs



Analysis

The 1-Y-T-Bill and the overnight rate follow similar trend. Their graphs go in the same direction. Vaguely putting, they both start off high in 2000, dip around 2002-03, then rise consistently till the end of 2007 and then dip sharply due to the GFC and then increase consistently.

As far as the inflation rate goes, it was consistently the same till 2008, had a sharp dip, hit rock bottom in 2009, and then rose consistently till 2011 and then has been decreasing.

The CDS Spread was 100 in 2000 and then shows a decreasing trend from 2000 till 2007 and then it soared in 2008. It sky rocketed. It was 200 in 2008 and it peaked at 500 in 2009. After that, it has been consistently declining. A higher CDS spread indicates that the market is more concerned about the risk of default. After the bubble burst, people anticipated even more people to default on their mortgage paymts which led them to shorting it via CDS.

The inflation rate dipped around the time of GFC indicating that the economy was performing poorly.

The overnight rate is used as a monetary tool by the BOC to influence the activity in the economy. It was lowered significantly since 2007. It was done taking into account the aftermath of GFC. Since the economy was slowing down, the overnight rate was reduced strikingly (it hit rock bottom at 0.25% in 2009) to make it easier for banks and in turn people to borrow in order to induce them to spend more.

Conclusion

The GFC had a negative impact on the Canadian housing market. Home prices fell by about 15% from their peak in 2008. This led to a decline in household wealth and a decrease in consumer spending.

The Canadian government responded to the GFC with a number of measures, including:

- A fiscal stimulus package of \$63 billion
- A reduction in interest rates by the Bank of Canada
- The purchase of mortgage-backed securities by the government

These measures helped to mitigate the impact of the GFC on the Canadian economy. However, the recovery was slow and uneven. The unemployment rate did not return to its pre-crisis level until 2013.

The GFC was a major economic shock that had a significant impact on Canada. The government's response to the crisis helped to mitigate the damage, but the recovery was slow and uneven. The long-term effects of the crisis are still being felt today like the reduced manufacturing sector and the widening gap between the rich and the poor.

References

1. <https://fred.stlouisfed.org/series/TB1YR>
2. [Policy interest rate - Bank of Canada](#)
3. [Consumer price index - Bank of Canada](#)
4. [Investopedia](#)